SECOND YEAR (Computer Science) ...RING & TECHNOLOGY

SPRING SEMESTER EXAMINATIONS 2024

Time : 3 Hours

Batch 2022

Dated : 30-JUL-24

Max Marks : 60

Computer Architecture & Organization - CS-252

Ms. Sabika Raza

Instructions:

- Attempt all the questions.
- Keep your answers brief and specific.

Q1 Describe the following question? Each question contain equal marks 20 marks **(C2, PLO-1)**

(a)What is the significance of the cache hit ratio, and how can it be improved?

(b)Explain the role of general-purpose registers in a CPU. How do they differ from special-purpose registers?

(c) Explain the RISC Architecture in detail with the help of a diagram. State the advantages of RISC over CISC approach?

(d) Explain the Harvard Architecture. State the difference between Harvard and Vonn Neumann architecture

Q2 Differentiate between them 20 marks **(C2, PLO-1)**

(a)Describe the differences between direct, associative, and set-associative cache mapping techniques

(b)Difference between segment, general and special purpose register

(c)Compare and contrast LRU and FIFO replacement algorithms. Under what circumstances might one be preferred over the other?

(d) Differentiate between zero and three addressing instruction format

Q3 Solve them 20 marks **(C3,PLO-3)**

(I) The content of following register are

CS = 3344h
DS = 1212h
SS = 4456h
BX = 1111h
BP = 8467h
IP = 2345h

Calculate the corresponding physical addresses for the address bytes in CS,DS AND SS

(II) Evaluate the following expression using all the four instruction formats:

$$X= (A+B+C) + [D/(E-F)]$$

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Seat No. CR-22049.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR (ARTIFICIAL INTELLIGENCE/ DATA SCIENCE/ CYBER SECURITY)
SPRING SEMESTER EXAMINATIONS 2024
Batch - 2022

Time: 3 Hours | **Dated:30-07-2024**
Max.Marks:60

COMPUTER ARCHITECTURE & ORGANIZATION - CS-252

Instructions:

- **Attempt all the questions. All questions carry equal marks.**
- **Keep your answers brief and specific.**

Q1 a) Explain the RISC Architecture in detail with the help of a diagram. State the advantages of RISC over CISC approach? **[5] (C2, PLO1)**

Q1 b) Describe the major functions of an I/O module with the help of a block diagram. Also, name the three techniques used by I/O module for basic I/O operations. **[5] (C2, PLO1)**

Q1 c) Discuss the function of following registers: **[2] (C2, PLO1)**

i. IR
ii. PC
iii. MDR
iv. MAR

Q2 a) What do you understand by the term cache mapping. Briefly discuss the replacement algorithms associated with cache mapping techniques. **[4] (C2, PLO1)**

Q2 b) Explain the basic function of a DMA Module. What is the advantage of DMA over Programmed and Interrupt Driven I/O? **[4] (C2, PLO1)**

Q2 c) Draw the Harvard Architecture. State the difference between Harvard and Vonn Neumann Architecture? **[4] (C2, PLO1)**

Q3 a) The contents of the Code Segment, Extra Segment and Stack Segment registers are as follows:

i) CS = 1280h
ii) ES= 4111h
iii) SS= 5353h

Calculate the corresponding effective addresses for the address bytes in CS, ES and SS, given the values of DI, IP and SP are 1001h, 2564h and 1256h respectively. **[4] (C3, PLO3)**

Q3 b) What is meant by instruction format? Describe the most common fields in an instruction format. **[4] (C3, PLO3)**

Q3 c) Describe the following addressing modes with two examples of each: **[4] (C3, PLO3)**

i. Register Direct Addressing Mode
ii. Immediate Addressing Mode

P.T.O

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SEAT NO. CT-21090

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
SECOND YEAR (BS IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2023
Batch 2021

Time : 3 Hours | **Dated : 10-AUG-23**
Max Marks : 60

Computer Architecture & Organization - CS-252

Q1) /15 marks (C3,PLO3)

a) Explain the following; (5 marks)
   i) Opcode ii) Address Field iii) Mode

b) What are the functions of Load & Store and in which instruction format they are found.(5 marks)

c) Name the 4 instruction formats with example. (5 marks)

Q2) /15 marks (C3,PLO3)

a) Solve X=A*B + C*C using zero address instruction. (5 marks)

b) Solve X=(A+B) * (C+D) using two address instruction. (5 marks)

c) Solve Y=(A-B) / [C+(D*E)] using three address instruction. (5 marks)

Q3) /15 marks (C3,PLO3)

a) Name the addressing mode: (5 marks)
   i) Add r4 ,hashtag 3 – Immediate
   ii) Add r3 , r4 Register Indirect
   iii) Load r1 , (10011) Direct.
   iv) Mov ax , [bx] Index
   v) Mov ax , [bx+100H] Index plus base

b) Explain Register Indirect Addressing Mode with the help of 3 examples. (5 marks)

c) Write 4 limitations of Register Direct Addressing Mode. (5 marks)

Q4) /15 marks(C2,PLO1)

a) Write difference between fixed program computer and stored program computer. (5 marks)

b) Draw RISC & CISC architectures. (5 marks)

c) Explain the following: (5 marks)
   i) PC ii) MAR iii) MDR iv) AC

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Information Technology

Midterm Examination Spring-2021 (Second Year BS-CSIT)

Time: 90 minutes | Computer Architecture and Organization (CS-252) | Dated: **25-06-2021** Max Marks: 20

Note: Attempt all (five) questions.

Q-1) what is computer architecture and how it differs from organization? Also explain classical components of computer.

Q-2) Elaborate the difference between Harvard architecture and Von Neumann architecture?

Q-3) Differentiate between Dual core vs Quad core processors.

Q-4) what is a bus? Explain Address bus, Data bus and Control bus with respect to Intel 8085 architecture.

Q-5) Draw and explain Instruction cycle state diagram.

SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2021
BATCH 2019

Time: 3 Hours | **Dated:31-08-2021** | **Max.Marks:60**

Computer Architecture & Organization - CS-252

Note: Attempt all questions.

Q-1-a) How does computer architecture differ from computer organization? (6)

Q-1-b) Differentiate between Von Neumann and Harvard architecture. Also, explain Moore's law. (6)

Q-2-a) Draw the block diagram of the Intel 8085 CPU. Explain control and status signals of 8085 CPU. (6)

Q-2-b) What is cache? And what are its types? also, discuss memory interleaving. (6)

Q-3-a) Draw and explain the instruction cycle state diagram. How many types of addressing modes are for the instruction sets? explain any four (4) of them. (6)

Q-3-b) Draw and explain memory hierarchy levels? also, discuss the characteristics affecting it. (6)

Q-4-a) Discuss the main process schedulers briefly in OS? What is pre-emtive scheduling? (6)

Q-4-b) What are the kernel and its types in an OS? also, differentiate between firmware and OS. (6)

Q-5-a) Add the following numbers using 2's complement: (4)

i) 45_{10} & -28_{10} ii) 25_{10} & -55_{10}

Q-5-b) What is the thread? Its types? and advantages over the process in OS? (6)

Q-5-c) How many address lines are required to address memory of 1450Kb? (2)